

Code Layout, Readability and Reusability

Project Name

Smart Lender – Loan Eligibility Prediction System

1. Introduction

The Smart Lender project follows a modular and well-organized code structure to ensure readability, maintainability, and reusability. The project separates machine learning logic, web application components, datasets, templates, and documentation into dedicated folders. This approach simplifies development, debugging, testing, and future enhancements.

2. Project Directory Structure

Smart Lender

```
|
|
|— Dataset/
|   |— loan_prediction.csv
|   └— generate_dataset.py
|
|— Training/
|   |— train_model.py
|   └— eda_outputs/
|
|— model/
|   |— saved_model.pkl
|   └— scaler.pkl
|
|— templates/
|   |— home.html
|   |— predict.html
|   └— submit.html
|
|— app.py
|— requirements.txt
```

└─ README.md

└─ .gitignore

3. Code Readability

The project follows coding practices that improve readability:

- Meaningful variable and function names.
 - Proper indentation and formatting.
 - Logical separation of frontend, backend, and machine learning modules.
 - Clear comments for important sections.
 - Consistent coding style throughout the project.
-

4. Code Reusability

The Smart Lender project is designed to encourage code reuse.

Examples include:

- The trained machine learning model is stored as a .pkl file and reused for every prediction.
 - Preprocessing logic can be reused for future datasets.
 - Flask routes are organized to avoid duplicate code.
 - HTML templates are reusable and easy to modify.
-

5. Modular Design

The project is divided into independent modules:

Module	Responsibility
Dataset	Stores training dataset
Training	Model training and evaluation
Model	Saved machine learning model
Templates	User interface pages

Flask Application Handles user requests and predictions

6. Best Practices Followed

- Modular project structure
 - Separation of concerns
 - Reusable machine learning model
 - Version control using Git and GitHub
 - Easy maintenance and scalability
-

7. Advantages

- Easy to understand
 - Simple to maintain
 - Faster debugging
 - Improved collaboration
 - Supports future feature additions
-

8. Conclusion

The Smart Lender project follows a clean and modular code layout that enhances readability, maintainability, and reusability. The structured organization enables efficient development while supporting future improvements and scalability.